

Types of Liver Diseases — Reference Guide

A detailed reference describing the major types of liver disease, their causes, symptoms, diagnosis, and treatment. It is intended as background knowledge for a liver-health information system.

Medical disclaimer: This document is an educational reference compiled for general information and for use in a retrieval-augmented (RAG) knowledge base. It is not a substitute for professional medical advice, diagnosis, or treatment. Always consult a qualified physician, hepatologist, or registered dietitian before making decisions about diet, medication, or lifestyle. In an emergency, contact local emergency services immediately.

1. Overview

Liver diseases can be broadly grouped into infections (viral hepatitis), fat-related disease (fatty liver), alcohol-related disease, immune and genetic disorders, scarring (fibrosis and cirrhosis), and tumours. Many conditions, if untreated, progress along a common path: inflammation → fibrosis → cirrhosis → liver failure or cancer.

2. Fatty Liver Disease

2.1 Non-Alcoholic Fatty Liver Disease (NAFLD / MASLD)

Fat builds up in liver cells in people who drink little or no alcohol. It is now the most common liver disease worldwide and is strongly linked to obesity, diabetes, and metabolic syndrome. It is increasingly called MASLD (metabolic dysfunction-associated steatotic liver disease).

- Causes: obesity, type 2 diabetes, insulin resistance, high cholesterol, poor diet.
- Symptoms: often none; sometimes fatigue or mild right-upper-abdomen discomfort.
- Progression: simple fat can advance to NASH (inflammation), then fibrosis and cirrhosis.
- Diagnosis: ultrasound, FibroScan, LFT, and sometimes biopsy.
- Treatment: weight loss, diet, exercise, and control of diabetes and cholesterol.

2.2 Alcoholic Fatty Liver Disease

The earliest stage of alcohol-related liver damage, caused by fat accumulation from excess drinking. It is often reversible if alcohol is stopped early.

3. Viral Hepatitis

Hepatitis means inflammation of the liver. Five main viruses cause it: A, B, C, D, and E.

Type	Spread	Nature	Prevention
Hepatitis A	Contaminated food/water	Acute, usually self-limiting	Vaccine, hygiene
Hepatitis B	Blood, sexual contact, mother-to-child	Acute or chronic	Vaccine, safe practices
Hepatitis C	Blood contact, needles	Often chronic	No vaccine; curable with antivirals
Hepatitis D	Only with hepatitis B	Worsens hepatitis B	Hepatitis B vaccine
Hepatitis E	Contaminated water	Acute; risky in pregnancy	Clean water, hygiene

- Common symptoms: jaundice, fatigue, nausea, loss of appetite, dark urine, abdominal pain.
- Chronic hepatitis B and C are major causes of cirrhosis and liver cancer.
- Hepatitis B and C can now be well controlled or cured with modern antiviral drugs.

4. Alcoholic Liver Disease (ALD)

A spectrum of damage from long-term heavy alcohol use, ranging from fatty liver to alcoholic hepatitis and finally cirrhosis.

- Alcoholic fatty liver: reversible early stage.
- Alcoholic hepatitis: liver inflammation that can be mild or life-threatening.
- Alcoholic cirrhosis: permanent scarring from years of drinking.
- Main treatment: complete and permanent alcohol cessation, nutrition, and medical care.

5. Liver Fibrosis and Cirrhosis

5.1 Fibrosis

Fibrosis is the build-up of scar tissue as the liver tries to repair ongoing damage. Early fibrosis can improve if the cause is removed.

5.2 Cirrhosis

Cirrhosis is advanced, widespread scarring that permanently affects liver structure and function. It is the end result of many chronic liver diseases.

- Causes: chronic hepatitis B/C, long-term alcohol use, advanced fatty liver, autoimmune and genetic diseases.
- Symptoms: fatigue, jaundice, fluid in the abdomen (ascites), leg swelling, easy bruising, and confusion in advanced stages.
- Complications: internal bleeding from varices, liver cancer, and liver failure.
- Treatment: treat the cause, manage complications, and consider transplant when severe.

6. Autoimmune and Cholestatic Liver Diseases

- Autoimmune hepatitis: the immune system attacks liver cells; treated with immune-suppressing drugs.
- Primary Biliary Cholangitis (PBC): slow immune-mediated damage to small bile ducts.
- Primary Sclerosing Cholangitis (PSC): scarring and narrowing of bile ducts, often linked to inflammatory bowel disease.

7. Genetic and Metabolic Liver Diseases

- Hemochromatosis: the body absorbs and stores too much iron, damaging the liver.
- Wilson's disease: copper accumulates in the liver and brain.
- Alpha-1 antitrypsin deficiency: a genetic protein defect that can injure the liver.

8. Liver Cancer

The most common primary liver cancer is hepatocellular carcinoma (HCC), which usually develops on a background of cirrhosis or chronic hepatitis. The liver can also be affected by cancers that spread from other organs (metastatic cancer).

- Risk factors: cirrhosis, chronic hepatitis B/C, heavy alcohol use, advanced fatty liver.
- Symptoms: weight loss, abdominal pain or mass, worsening jaundice, loss of appetite.
- Screening: ultrasound (often with AFP blood test) every 6 months in high-risk patients.
- Treatment: surgery, ablation, targeted therapy, or transplant depending on the stage.

9. Other Liver Conditions

- Drug-Induced Liver Injury (DILI): damage from medicines, overdoses, or supplements.
- Liver abscess: a pus-filled infection, often bacterial or amoebic.
- Liver cysts and hemangiomas: usually benign fluid- or blood-filled growths.
- Gallstone-related disease: stones can block bile ducts and injure the liver.
- Acute liver failure: rapid loss of liver function, a medical emergency.

10. How Liver Diseases Are Diagnosed

- Blood tests: LFT, viral markers, autoimmune and genetic panels, AFP.
- Imaging: ultrasound, FibroScan, CT, and MRI.
- Liver biopsy: a tissue sample when the diagnosis or stage is unclear.

11. General Principles of Treatment

- 1 Identify and treat the underlying cause (virus, alcohol, fat, immune, or genetic).
- 2 Stop all liver toxins, especially alcohol and unnecessary drugs.
- 3 Support the liver with good nutrition and weight control.
- 4 Monitor for and manage complications early.
- 5 Refer advanced cases to specialist and transplant centres.

Summary: liver diseases range from easily reversible fatty liver to life-threatening cirrhosis and cancer. Early detection, removing the cause, and regular follow-up are the keys to good outcomes across almost every type.